Press Line — Machine Profile Sheet

Machine Profile Sheet · Quick Reference

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Owner	Manufacturing Engineering
Applies To	Press Line — Plant A (Pune), Plant C (Ahmedabad); units Press_Line_1–20

1. Capacity

Metric	Value
Fleet size (this machine type)	20 units
Average ideal cycle time	1.16 min/part
Theoretical rated output	52 parts/hour — 415 parts/shift (480 min, no losses)

2. Rated Speed / Mechanical Specification

Parameter	Specification
Rated tonnage	150 – 400 tons
Stroke rate	8 – 20 strokes/min
Hydraulic system pressure	180 – 210 bar nominal
Die clamping	Hydraulic quick-clamp, 4-point
Safety system	Light curtain + two-hand anti-tie-down control

3. Typical OEE Target

Industry world-class benchmark for discrete manufacturing is 85% OEE (90% availability × 95% performance × 99% quality). Measured performance for this machine type across the current dataset is **89.1% OEE** (availability 95.6%, performance 95.3%, quality 97.8%). Plant target for this machine type is set at ≥85% OEE, with any sustained reading below 80% triggering a reliability review under Plant_Maintenance_Policy.

4. Critical Components

- Hydraulic power unit (pump, valves, accumulator) — dominant failure point, see DT-01.
- Ram cylinder seals — primary source of pressure-loss failures.
- Die clamp mechanism — drives changeover time (DT-03).
- Hydraulic hoses and fittings — highest-frequency leak point, weekly inspection required.

5. Common Failure Modes (ranked by share of logged downtime)

Failure Mode	Code	Share of Downtime	Avg Duration
Hydraulic System Failure	DT-01	58%	131 min avg
Raw Material / Feed Shortage	DT-02	28%	52 min avg
Tooling / Die Changeover Delay	DT-03	10%	34 min avg
Spindle / Main Bearing Failure	DT-08	5%	46 min avg